

MIPS chip architecture goes open source

Wave Computing has announced that it will open source its MIPS instruction set architecture (ISA) to accelerate the ability for semiconductor companies, developers and universities to adopt and innovate using MIPS for next-generation system-on-chip (SoC) designs.

The MIPS ISA will be open sourced providing full access to both the 32-bit and 64-bit versions, free of charge – with no licensing or royalty fees. This is being done under the MIPS Open programme launched by the Silicon Valley company. Under the programme, participants will have access to existing MIPS patents as well.

The company said that the details on open sourcing MIPS and information on how to participate will be made available in the first quarter of 2019.

Wave acquired MIPS Technologies in June 2018. Since the time of acquisition, Wave CEO Derek Meyer, a MIPS veteran, has been planning on going open source – as revealed to *EE Times* by Art Swift, president, Wave Computing's MIPS IP business.

The MIPS Open initiative is a key part of Wave's 'AI for All' vision, according to Lee Flanagan, Wave's senior vice president and chief business officer. This initiative will be moderated by Wave Computing and an advisory committee, which will include industry-leading OEMs, partners, universities and technology luminaries who will help guide community-driven innovations on MIPS.

In addition, certified verification partners will assist in ensuring compatibility of implementations and preventing architectural fragmentation.

Lee believes that the MIPS-based solutions developed under MIPS Open will complement their existing and future MIPS IP cores.

"This will ensure that current and new MIPS customers will have a broad array of solutions from which to choose for their SoC designs, and will also have access to a vibrant MIPS development community and ecosystem," he asserts.

"We hope that Trill and IStreamable will provide a strong foundation for streaming or temporal processing for current and future open source offerings," writes James Terwilliger, principal software engineer, Microsoft Azure, in a blog post.

To try Trill, Microsoft is inviting people to reach out to the company via email, using asktrill@microsoft.com.

An open source analytics tool to know how customers use a website

Microsoft's search engine Bing has announced the beta release of Clarity, a new analytics product that lets site owners see how users interact on their websites.

The free, open source tool will help site owners decide on what exactly they should change and improve on their sites to optimise conversion, engagement and retention.



Clarity allows site owners to replay users' interactions such as mouse movements, touch gestures and click events. By doing so, Web masters can get a deeper understanding of user behaviour, their pain points and requirements, to build a more compelling and user friendly website.

It also has other typical functionalities such as a heat map and scroll map.

Clarity respects users' privacy by using text masking. Text is masked by default in the instrumentation layer and is not uploaded.

In a blog post, the Clarity team claims that websites of all sizes can benefit from the tool.

The tool works on any HTML Web page (desktop or mobile) after adding a small snippet of JavaScript to the website.

Site owners can avail the tool using their Microsoft accounts. As it is still in the beta stage, when you create a new project, it will be added to the waitlist. Once your project is approved for onboarding, you can get the code and add it to your website to use Clarity.

One early tester of Clarity is the food blog *Cook with Manali*. While using Clarity, the site owner noticed that a lot of users were abandoning the page before reaching the bottom, which has the most important information about the recipe.

After replaying sessions of users who abandoned her blog, the site owner could figure out that those who were only interested in the recipe, which is at the bottom, scrolled through the long post, gave up midway and abandoned the page. Based on that finding, she added a 'Jump to recipe' button at the top of the page, giving users a faster path to the recipe content.